ACS-2906-001 St. No- 3169763

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1 a) $-51 \div 6$

$-51 \div 6 = -8.5$

$51 = 1\times32 + 1\times16 + 0\times8 + 0\times4 + 1\times2 + 1\times1$

$= 32 + 16 + 0 + 0 + 2 + 1$

$= 48 + 2 + 1$

$= 51$

$(51)_{10} = (110011)_2$

$6 = 1\times4 + 1\times2^1 + 0\times2^0$

$= 4 + 2 = 6$

$(6)_{10} = (110)_2$

```
        1000.1
110 | 110011
     -110
       00
        -0
        01
         0
         11
         -0
         110
         110
          0
```

Convert binary into decimal =

$1000.1 = (1\times2^3) + (0\times2^2) + (0\times2^1) + (0\times2^0) + (1\times2^{-1})$

$= 8 + 0 + 0 + 0 + 0.5 = 8.5$

The magnitude is correct
we just need to add negative sign
Final Answer →
-1000.1

2 Total bit = 8 $n = 4$
Sign bit = 1 bias = 3
$k = 3$

(a) $S = 0$, $e = 000$ frac = 0001.

Bits = 0 000 0001.

(b). Exponent fields is 000
fraction = denormalized form.

(c) Sign = 0 (positive)
fraction = 0001
Which is equal to 0.0001 in decimal

Bias = 3

$$\text{Value} = (-1)^S \times M \times 2^{(1-\text{bias})}$$
$$= (-1)^0 \times 0.0001 \times 2^{1-3}$$
$$= 1 \times 0.0001 \times 2^{-2}$$
$$= 2^{-4} \times 2^{-2}$$
$$= 2^{-6}$$

d) The largest non-infinite number is the normalized number.

→ Sign S: 0 (+ve)

→ E: 111 (for infinite and NAN)

so we use

$$(110)_2 = (1\times 2^2)+(1\times 2^1)+(0\times 2^0)$$
$$= 4+2+0 = (6)_{10}.$$

→ fraction $= (1111)_2$

The formula for Normalized.

$$\underline{\text{value}} = (-1)^S \times (1 + \text{fraction}) \times 2^{E-\text{Bias}}$$

$$\text{fraction} = 1.1111$$
$$= 1 + 1\times 2^{-1} + 1\times 2^{-2} + 1\times 2^{-3} + 1\times 2^{-4}$$

$$= \frac{16+8+4+2+1}{16} = \frac{31}{16}.$$

$$(-1)^0 \times \frac{31}{16} \times 2^{6-3}$$

$$1 \times \frac{31}{16} \times 2^3 = \frac{31}{2}$$

Value = 15.5

3. a) -45.625

Sign bit : 1 (because number is -ve)

Converting 45.625 into binary

$45 = (1\times2^5) + (0\times2^4) + (1\times2^3) + (1\times2^2) + (0\times2^1) + (1\times2^0)$

$= (101101)_2$

fractional part → (0.625)

$0.625 \times 2 = 1.25$ (1)

$0.25 \times 2 = 0.5$ (0)

$0.5 \times 2 = 1$ (1)

$(101)_2$

$45.625 = (101101.101)_2$

Normalize the Binary Number →

$101101.101 = 1.01101101 \times 2^5$

$E = 127+5 = 132$

$132 = 128+4 = 2^7 + 2^2 = 10000100$

$E = 10000100$

fractional field → 01101101000000000000000

$-45.625 = 1 \quad 10000100 \quad 01101101000000000000000$

b) 0.1

S = 0 (+ve)

fractional part →

$0.1 \times 2 = 0.2$ (0)

$0.2 \times 2 = 0.4$ (0)

$0.4 \times 2 = 0.8$ (0)

$0.8 \times 2 = 1.6$ (1)

$0.6 \times 2 = 1.2$ (1)

$0.2 \times 2 = 0.4$ (0)

the pattern 0011 repeats

$(0.1)_{10} = 0.0001100110011001100110011\ldots$

Normalize: to get 1 on the left of the point

$1.10011001100110011\ldots \times 2^{-4}$

$p = -4$

Exponent part

$127 + (-4) = 123$

$123 = 64 + 32 + 16 + 8 + 2 + 1 = 0111011$

E = 0111011

Fractional field = 10011001100110011001101 (rounded)

Final Answer

0 0111011 10011001100110011001101

4. R1 : 010001010000 R2: 010000100000.

R1 + R2

Bias = $2^{e-1}-1 = 2^{5-1}-1 = 2^4-1 = 16-1 = 15$

R1

S = 0. (+ve)

Exp 10001 = 17

Fraction = 010000 + 1 = 1.010000.

E = exp - bias = 17-15 = 2

fraction = 1.010000×2^2

R2 =

S = 0 (+ve)

exp = 10000 = $1 \times 2^4 = 16$

E = exp - bias = 17-16 = 1

fraction = 1 + 100000 = 1.100000

= 1.100000×2^1

1.010000×2^2 ← Matching Exponent

1.100000×2^1

$$\begin{array}{r} 1.010000 \times 2^2 \\ + \quad 0.110000 \times 2^2 \\ \hline 10.000000 \times 2^2 \end{array}$$

$10.000000 \times 2^2 = 1.000000 \times 2^3$

Actual exponent = 3

we need to write this in 12 bit format

S = 0 (+ve)

E = bias + p = 15 + 3 = 18

18 = 16 + 2 = $(10010)_2$

E = 10010.

fraction = $(000000)_2$

final Answer →

0 10010 000000.